

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

 show ip router ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? s0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

 R3 is linking to R2 via R1, not direct connect to R2, the traffic uses a longer path to link R3 and R2.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

 router ospf 1

 network 192.168.23.0 0.0.0.3 area 0

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? s0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

 64 + 1 = 65 (R3 s0/0/1 + R2 G0/0 cost)

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

Please ensure reference bandwidth is consistent across all routers.

Why would you want to change the OSPF default reference-bandwidth?

To make sure the cost calculations accurate. This can prevent OSPF choosing another path or Ethernet that is faster automatically.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0 : cost = $781 + 1 = 782$ (R1 s0/0/1 cost + R3 G0/0 cost)

192.168.23.0 : cost = $781 + 64 = 845$ (R1 cost + R3 s0/0/1 cost)

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562, as cost = $781 + 781 = 1562$ (R1 + R3)

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

As the cost of the direct connection to R3 is higher than the path through R2, therefore the path is changed.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

For each router in the OSPF routing domain to be unique, you need to control the router ID. This stops ID conflicts and makes sure that path calculations and the DR/BDR election process work as expected.

2. Why is the DR/BDR election process not a concern in this lab?

The topology mainly uses serial point-to-point links and LAN segments where there is only one router. Only on multi-access networks like Ethernet, where more than one OSPF router is connected to the same segment, do DR/BDR elections happen.

3. Why would you want to set an OSPF interface to passive?

As it can prevent the transmission of OSPF Hello packets and routing updates out of that interface. This can reduce computational costs and improves security.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The use of Areas allows OSPF to be hierarchical. It can reduce the size of the Link State Database on each router, enable the network to scale significantly.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

Network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route, as the decision is based on the administrative distance and AD of EIGRP is 90 which is the smallest. The metric does not matter if the administrative distances are not the same